

GENERAL SPECIFICATION G7.1

## Air Conditioning Equipment

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## G7.1 AIR CONDITIONING EQUIPMENT

### G7.1.1 Energy Efficiency Specifications

- (a) The contractor shall supply air conditioning system and equipment to meet the following:
  - (i) Minimum DSE (kW/RT):  $\eta_t$  ( $\eta_c$ ,  $\eta_a$ ) where  $\eta_t = \eta_c + \eta_a$ ;
  - (ii)  $\eta_t$  = Total System Efficiency;
  - (iii)  $\eta_c$  = System kW/ton excluding the air distribution equipment;
  - (iv)  $\eta_a$  = Air Distribution equipment kW/ton; and
  - (v) And where ( $\eta_c$ ,  $\eta_a$ ) shall meet their respective thresholds.
- (b) Air Cooled Chilled Water System/Unitary Air Conditioning System
  - (i) Minimum DSE (kW/RT) : 1.03 (0.78, 0.25).
- (c) Water Cooled Chilled Water System
  - (i) Minimum DSE (kW/RT) : 0.93 (0.68, 0.25) for <500RT; and
  - (ii) and DSE (kW/RT) : 0.90 (0.65, 0.25) for >500RT.

### G7.1.2 Acts, Regulations and By Laws

- (a) The ACMV system shall be designed and carried out to comply with all relevant acts, regulations and by laws, and all amendments thereof, in particular the following:
  - (i) Building Control Act (Chapter 29);  
Building Control Regulations;
  - (ii) Fire Safety Act (Chapter 109A);  
Fire Safety (Building Fire Safety) Regulation;
  - (iii) Public Utilities Act (Chapter 261)  
Public Utilities (Water Supply) Regulations;
  - (iv) Electricity Act (Chapter 89A);  
Electricity (Electrical Installation) Regulations;
  - (v) Environmental Public Health Act (Chapter 95);
  - (vi) Environmental Protection & Management Act (Chapter 94A); and
  - (vii) Workplace Safety and Health Act (Act 7 of 2006).

### G7.1.3 Reference Standards

- (a) The system shall be designed and installed with reference to the following codes and standards:

| Standard             | Subject                                                                               |
|----------------------|---------------------------------------------------------------------------------------|
| AHRI 1060            | Performance rating of air-to-air exchangers for energy recovery ventilation equipment |
| AHSRAE Standard 62.1 | Ventilation for acceptable indoor air quality                                         |

| Standard             | Subject                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE Standard 111  | Measurement, testing, adjusting, and balancing of building HVAC systems                                                                            |
| ASHRAE Standard 15   | Safety standard for refrigeration systems                                                                                                          |
| ASHRAE Guideline 0   | The commissioning process                                                                                                                          |
| ASHRAE Guideline 1.1 | HVAC & R technical requirements for the commissioning process                                                                                      |
| BS 476-12            | Fire tests on building materials and structures, Part 12: Method of test for igniability of products by direct flame impingement                   |
| SS CP 5              | Code of practice for electrical installations                                                                                                      |
| SS CP 83-2           | Code of practice for constuctions computer-aided design (CAD), Part 2 : CAD symbols                                                                |
| SS 212               | Specification for aluminium alloy windows                                                                                                          |
| SS 299-1             | Specification for fire resistant cables, Part 1 : Performance requirements for cables required to maintain circuit integrity under fire conditions |
| SS 530               | Code of practice for energy efficiency standard for building services and equipment                                                                |
| SS 553               | Code of practice for air Conditioning and Mechanical Ventilation in Buildings                                                                      |
| SS 554               | Code of practice for indoor air quality for air-conditioned buildings                                                                              |
| SS 591               | Code of practice for long term measurement of central chilled water system energy efficiency                                                       |

- (i) ASHRAE Handbook of Fundamentals.
- (ii) Code of Practice for the Control of Legionella Bacteria in Cooling Towers.
- (iii) Code of Practice for Fire Precautions in Buildings.
- (iv) Code of Practice on Environmental Health.

#### **G7.1.4 Chiller**

##### **(a) Refrigeration Machines**

- (i) Hermetic refrigeration machines shall be supplied and installed generally as shown in the Schematic Drawings. The machines shall provide cooling capacity as shown in the Particular Specification P7 – Building Services and schematic drawings. It shall be capable of working at reduced load conditions, down to 10% of its specified cooling capacity under specified operating condition.

- (ii) Chiller Optimisation Program should balance out the load between operating Chillers to allow partial loading of chillers for maximum efficiency.
  - (iii) Chillers shall be supplied and operated with refrigerants which have an Ozone Depletion Potential (ODP) of zero or Global Warming Potential (GWP) of less than 100. CFC and HCFC refrigerants will not be acceptable. The type of refrigerant shall be of non-CFC refrigerant (e.g. HFC134a, HCFC-22, HCFC 123, etc.) and suitable for installation in building as approved by the local Authorities.
  - (iv) Each refrigeration machine shall be supplied complete with compressor, refrigerant condenser and evaporator, pump out system, prime mover, oil pumps, purge system (if required) and control system.
  - (v) The overall heat transfer coefficients shall allow for water chiller scale factor not less than 0.0001 and refrigerant condenser scale factor not less than 0.0002 for fouling of the water side of the tubes (scale factor expressed as sq m hr degree Celsius/Kcal).
  - (vi) The machines shall be arranged to allow any of the machines to lead in the sequence of operation as specified later.
  - (vii) The machines shall be arranged with connecting pipeworks isolating valves, which will allow any of the refrigeration machines to be isolated without interrupting the operation of the other machines.
  - (viii) Automatic Tube Cleaning System (sponge ball type) shall be provided for the heat exchangers.
- (b) Centrifugal Refrigeration Machines Using Low Pressure Refrigerant
- (i) Compressor
    - 1) The volute casings shall be cast from high quality close grain grey cast iron or suitable semi-steel.
    - 2) The shaft shall be an alloy steel forging and impellers shall be statically and dynamically balanced and the entire rotating assembly must operate well below its first critical speed.
    - 3) Bearings shall be fully lubricated and each machine shall incorporate an oil storage tank with oil pump (electrically operated), which is to supply the lubricating system. A cooling coil and immersion heater shall be incorporated within the oil tank, both of which shall be automatically controlled to maintain the oil at the correct temperature. The oil lubrication system shall be interlocked with the compressor starting gear to ensure that the refrigeration plant cannot commence operation until the lubrication system is providing the correct quantity of oil, and at a satisfactory pressure to all bearings.
    - 4) The unit shall be complete with all necessary safety devices and automatic adjustments to the loads imposed on it. Capacity control shall be by means of variable speed drives which shall provide continuously variable capacity from 10% to 100% of rated compressor capacity under specified operating condition. The VSD shall be automatically controlled by a suitable chilled water control device to ensure a leaving chilled water temperature deviation of not more than +/- 0.5 degree Celsius.

(ii) Heat Exchangers

- 1) Heat exchangers shall be of the shell and tube pattern. The shell shall contain all necessary ports and flanges for steel pipe connections. Adequate access covers shall be provided for routine inspection and maintenance.
- 2) Heat exchangers shall have flanged end cover plates completely removable.
- 3) Allowance shall be made for any difference in the coefficient of expansion of the material used in the construction of the heat exchangers.
- 4) All necessary sight glasses and other accessories shall be provided.
- 5) The method of construction and the materials used shall be as recommended for these conditions of working. Special attention is drawn to the quality of the condenser cooling water. Heat exchangers shall be designed and pressure tested to the requirements of the manufacturer's and approved by S.O..
- 6) Water boxes shall be provided on the refrigerant condenser and evaporator to permit access to the tubes without disturbing the water connections. Water boxes shall be designed for 10.5 bar minimum working pressure.
- 7) Isolation valves shall be factory installed to allow isolation of refrigerant in the condenser.
- 8) The water chiller shall be provided with eliminators to ensure that liquid refrigerant is not carried over to the compressor.
- 9) Water pressure drop through the refrigerant condenser and evaporator shall not exceed 0.75 bar each, unless otherwise stated in the Schedule of Technical Requirements.
- 10) Water flow rate through the heat exchangers shall not exceed 3 m/s for both the condenser tubes and the chiller tubes.
- 11) Suitable tapping shall be provided in the water boxes for thermometers, control bulbs, gauges and sensing devices for water treatment.
- 12) Steel or approved division plates separating the water passes shall be easily removable for servicing.
- 13) Types and arrangement of filters shall be as recommended by the manufacturers for the satisfactory operation of the plant.

(iii) Purge Systems

- 1) Fixed purge units of the fully automatic type shall be provided for each compressor set to clear the system of any air and non-condensable gases.
- 2) Purge units shall be complete with all necessary piping and connections together with all associated controls, gauges and electrical wiring which shall be mounted local to each unit.

- 3) Oil separation equipment shall be provided comprising traps, strainers, floats, receivers, gauges, etc. and oil still or purifier as necessary.
  - 4) The purge unit shall also be capable of being manually operated.
- (c) Centrifugal Refrigeration Machines using Positive Pressure Refrigerant
- (i) Compressor
    - 1) The volute casings shall be cast from high quality close grain grey cast iron or suitable semi-steel.
    - 2) The shaft shall be an alloy steel forging and impellers shall be statically and dynamically balanced and the entire rotating assembly must operate well below its first critical speed.
    - 3) Bearings shall be fully lubricated and each machine shall incorporate an oil storage tank with oil pump (electrically operated), which is to supply the lubricating system or magnetic bearings. A cooling coil and immersion heater shall be incorporated within the oil tank, both of which shall be automatically controlled to maintain the oil at the correct temperature. The oil lubrication system shall be interlocked with the compressor starting gear to ensure that the refrigeration plant cannot commence operation until the lubrication system is providing the correct quantity of oil, and at a satisfactory pressure to all bearings.
    - 4) The unit shall be complete with all necessary safety devices and automatic adjustments to the loads imposed on it. Capacity control shall be by means of inlet guide vanes which shall provide continuously variable capacity from 10% to 100% of rated compressor capacity under specified operating condition. The inlet guide vanes shall be automatically controlled by a suitable chilled water control device to ensure a leaving chilled water temperature deviation of not more than +/- 0.5 degree Celsius.
  - (ii) Pump Out System
    - 1) A refrigerant pump out system complete with pump down compressor, piping, wiring, motor and storage vessel constructed in accordance with ASME Code for Unfired Pressure Vessels shall be installed on the chiller. The storage vessel shall be integral to the chiller but external to the condenser and evaporator shell and shall be large enough to hold the entire refrigerant charge.
  - (iii) Heat Exchangers
    - 1) Heat exchangers shall be of the shell and tube pattern. The shell shall contain all necessary ports and flanges for steel pipe connections. Adequate access covers shall be provided for routine inspection and maintenance.
    - 2) Heat exchangers shall have flanged end cover plates completely removable.
    - 3) Allowance shall be made for any difference in the coefficient of expansion of the material used in the construction of the heat exchangers.
    - 4) All necessary sight glasses and other accessories shall be provided.

- 5) The method of construction and the materials used shall be as recommended for these conditions of working. Special attention is drawn to the quality of the condenser cooling water. Heat exchangers shall be designed accordance to ASME Section VIII, Div. 1 and shall display an ASME nameplate which shows pressure and temperature rating. A pressure relief valve shall be installed on each heat exchanger.
  - 6) Water boxes shall be provided on the refrigerant condenser and evaporator to permit access to the tubes without disturbing the water connections. Water boxes shall be designed for 10.5 bar minimum working pressure.
  - 7) Isolation valves shall be factory installed to allow isolation of refrigerant in the condenser.
  - 8) The water chiller shall be provided with eliminators to ensure that liquid refrigerant is not carried over to the compressor.
  - 9) Water pressure drop through the refrigerant condenser and evaporator shall not exceed 0.75 bar each, unless otherwise stated in the Schedule of Technical Requirements.
  - 10) Water flow rate through the heat exchangers shall not exceed 3 m/s for both the condenser tubes and the chiller tubes.
  - 11) Suitable tapping shall be provided in the water boxes for thermometers, control bulbs, gauges and sensing devices for water treatment.
  - 12) Steel division or approved plates separating the water piazze shall be easily removable for servicing.
  - 13) Types and arrangement of filters shall be as recommended by the manufacturers for the satisfactory operation of the plant.
- (d) Rotary / Screw Refrigeration Machines using Positive Pressure Refrigerant
- (i) Compressor
    - 1) The compressor casings shall be cast from high quality close grain grey cast iron or a suitable semi-steel.
    - 2) The shaft and rotor shall be an alloy steel forging and shall be statically and dynamically balanced. The entire rotating assembly must operate well below its critical speed. A spring loaded check valve shall be installed at the compressor discharge outlet to prevent rotor backspin during shutdown.
    - 3) Bearings shall be fully lubricated and each machine shall incorporate an oil storage tank with oil pump (electrically operated), which is to supply the lubricating system. A cooling coil and immersion heater shall be incorporated within the oil tank, both of which shall be automatically controlled to maintain the oil at the correct temperature. The oil lubrication system shall be interlocked with the compressor starting gear to ensure that the refrigeration plant cannot commence operation until the lubrication system is providing the correct quantity of oil, and at a satisfactory pressure to all bearings.

- 4) The unit shall be complete with all necessary safety devices and automatic adjustments to the loads imposed on it. Capacity control shall be by means of slide valve which shall provide continuously variable capacity from 10% to 100% of rated compressor capacity under specified operating condition. The slide valve shall be automatically controlled by a suitable chilled water control device to ensure a leaving chilled water temperature deviation of not more than  $\pm 0.5$  degree Celsius.
- (ii) Pump Out System
- 1) A refrigerant pump out system complete with pump down compressor, piping, wiring, motor and storage vessel constructed in accordance with ASME Code for Unfired Pressure Vessels shall be installed on the chiller. The storage vessel shall be integral to the chiller but external to the condenser and evaporator shell and shall be large enough to hold the entire refrigerant charge.
- (iii) Heat Exchangers
- 1) Heat exchangers shall be of the shell and tube pattern. The shell shall contain all necessary ports and flanges for steel pipe connections. Adequate access covers shall be provided for routine inspection and maintenance.
  - 2) Heat exchangers shall have flanged end cover plates completely removable.
  - 3) Allowance shall be made for any difference in the coefficient of expansion of the material used in the construction of the heat exchangers.
  - 4) All necessary sight glasses and other accessories shall be provided.
  - 5) The method of construction and the materials used shall be as recommended for these conditions of working. Special attention is drawn to the quality of the condenser cooling water. Heat exchangers shall be designed accordance to ASME Section VIII, Div. 1 and shall display an ASME nameplate which shows pressure and temperature rating. A pressure relief valve shall be installed on each heat exchangers.
  - 6) Water boxes shall be provided on the refrigerant condenser and evaporator to permit access to the tubes without disturbing the water connections. Water boxes shall be designed for 10.5 bar minimum working pressure.
  - 7) Isolation valves shall be factory installed to allow isolation of the refrigerant in the condenser.
  - 8) The water chiller shall be provided with eliminators to ensure that liquid refrigerant is not carried over to the compressor.
  - 9) Water pressure drop through the refrigerant condenser and evaporator shall not exceed 0.75 bar each, unless otherwise stated in the Schedule of Technical Requirements.
  - 10) Water flow rate through the heat exchangers shall not exceed 3 m/s for both the condenser tubes and the chiller tubes.



- 11) Suitable tapping shall be provided in the water boxes for thermometers, control bulbs, gauges and sensing devices for water treatment.
  - 12) Steel or approved division plates separating the water passes shall be easily removable for servicing.
  - 13) Types and arrangement of filters shall be as recommended by the manufacturers for the satisfactory operation of the plant.
- (iv) Oil Separation System
- 1) Fixed oil separator of the fully automatic type shall be provided for each compressor set to remove oil in the refrigerant before entering the heat exchangers.
  - 2) Oil separator shall comprise traps, strainers, receivers, gauges, etc. and oil still or purifier as necessary.
  - 3) Means for removing oil trap in the heat exchangers shall be provided to ensure smooth and efficient operation.
- (v) Prime Mover
- 1) Each motor shall be of the squirrel cage type of sufficient capacity to efficiently fulfil the compressor brake horsepower requirements including gear drive losses. The electric motor shall be suitable for starting by auto-transformer starter or electronic soft starter. Motors and starters shall be to the approval of the relevant authorities.
  - 2) The gear drive and the electric motor shall be suitable for the type of service and entirely compatible with the compressor.
- (vi) Control Centre
- 1) Each refrigeration machine shall be complete with a microprocessor-based control centre, factory internally wired with replaceable modular component construction. The control centre shall include a liquid crystal display of minimum 16-line by 40 characters complete with function keys, start/stop button and alarm lights. The following minimum functions shall be provided at the control centre:
  - 2) Chilled water temperature controller with automatic control point adjustment.
  - 3) Compressor start/stop push button switches with indicator lights.
  - 4) Motor current control with maximum load selector switch for 15% to 100% full load current.
  - 5) Chilled water inlet and outlet temperature indication by LED display.
  - 6) Refrigerant suction and discharge temperature indication by LED display.
  - 7) Protective controls with red indicator lights and manual resets for high and low refrigerant pressures, low oil pressure, high oil temperature, chilled water and condenser water flow failures and with automatic reset for low chilled water temperature.

- 8) Dial pressure gauges for refrigerant evaporator pressure and condenser pressure, and compressor oil pressure. Refrigerant pressure gauges shall be supplied with equivalent saturation temperature scales in addition to pressure scales.
  - 9) Chilled water and condenser water flow switches to stop the compressor when chilled water or condenser water ceases to flow and to prevent the compressor being started until chilled water or condenser water flow has been established.
  - 10) Purge unit power switch (for low-pressure chiller).
  - 11) The control centre shall be equipped with software capability to perform totalling function and trend recording function. These functions shall have facilities for direct display on colour monitor and output to printer. The minimum function points to be included are listed as follows:
    - Chilled water supply and return temperature;
    - Chiller operating hours;
    - Evaporator refrigerant temperature;
    - Condenser water supply and return temperature;
    - Condenser refrigerant temperature;
    - Oil supply pressure;
    - Oil sump temperature; and
    - Percent motor rated load amps.
  - 12) The trend recording function shall be capable of retaining not less than 2000 records for each function point.
  - 13) A common colour monitor and printer shall be provided for hook-up to every chiller in each chiller plant room.
- (e) Reciprocating Water Chilling Units
- (i) Hermetic or semi hermetic reciprocating refrigeration machines shall be supplied and installed where required and shown in the Specification Drawings. The machines shall have a total capacity as shown on the Schedule of Technical Requirements and each machine shall be capable of working at reduced load of 25% or less of its specified cooling capacity.
  - (ii) Each refrigeration machine shall be complete with compressor, water chiller, prime mover, purge system and control system.
  - (iii) The overall heat transfer coefficients shall allow for water chiller scale factor not less than 0.0001 and refrigerant condenser scale factor not less than 0.0002 for fouling of the water side of the tubes. (Scale factor expressed as sq m hr degree Celsius/Kcal).

(iv) Details of other components are as described below:

1) Compressor

- The compressors shall be of the hermetic or semi-hermetic reciprocating type, direct coupled and complete with forged steel crank weights of one piece construction with integral counter weights to keep vibration to a minimum when operating at high speeds.
- The pistons shall be of lightweight construction with ground skirts and complete with compression and oil wiper rings.
- The cylinders shall be fitted with liners of the replaceable type.
- Lubrication system shall be of the forced feed type and compressors shall be fitted with gauge connections for reading oil, suction and discharge pressures and fitted with oil sight glass and cleanable soil filter.
- The unit shall be complete with all necessary safety devices and automatic adjustments to the loads imposed on it. Capacity control shall be by means of cylinder suction unloader, which shall provide a minimum of 2 stage capacity control for each compressor. The cylinder unloader shall be controlled by suitable chilled water control device to ensure a leaving chilled water temperature deviation of not more than  $\pm \frac{1}{2}$  degree Celsius.

2) Heat Exchangers

- Heat exchangers shall be of the shell and tube pattern. The shell shall consist of a fusion welded steel cylinder, with all necessary ports and flanges for steel pipe connections. Adequate access covers shall be provided for routine inspection and maintenance.
- Heat exchangers shall have flanged end cover plates completely removable.
- Allowance shall be made for any difference in the coefficient of expansion of the materials used in the construction of the heat exchangers.
- All necessary sight glasses and other accessories shall be provided.
- The method of construction and the materials used shall be as recommended by the manufacturer for these conditions of working. Special attention is drawn to the quality of the condenser cooling water. Heat exchangers shall be designed and pressure tested to the approval of the local Authorities.
- The water chiller shall be provided with eliminators to ensure that liquid refrigerant is not carried over to the compressor. Water pressure drop through the heat exchangers shall not exceed 0.75 bar of water.
- Types and arrangements of filters shall be as recommended by the manufacturers for the satisfactory operation of the plant.

3) Prime Mover

- Each hermetically sealed motor shall be of the squirrel cage type of sufficient horsepower to efficiently fulfil the compressor brake horsepower requirements including gear drive losses. Motors and starters shall be to the approval of the local Electricity Supply Authorities.

4) Control Centre

- Each refrigeration machine shall be complete with a control centre, factory internally wired and piped for mounting and connection to the system to include the following:
  - Chilled water temperature controller with automatic control point adjustment;
  - Compressor start/stop push button switches with indicator lights;
  - Chilled water inlet and outlet dial type thermometers;
  - Suction and discharge thermometers on refrigerant circuits;
  - Protective controls with red indicator lights and manual resets for high and low refrigerant pressures, low oil pressure, high oil temperature, chilled water and condenser water flow failures and with automatic reset for low chilled water temperature;
  - Dial pressure gauges for refrigerant evaporator pressure and condenser pressure, and compressor oil pressure. Refrigerant pressure gauges shall be supplied with equivalent saturation temperature scales in addition to pressure scales; and
  - Chilled water and condenser water flow switches to stop the compressor when chilled water or condenser water ceases to flow and prevent the compressor being started until the water flow has been established.

1) Insulation

The water chiller shell and water boxes, suction connection to the compressor and all other parts subject to condensation shall be insulated with an approved minimum 25 mm thick first quality factory applied insulation, complete with tough vapour proof skin.

2) Machinery Bases

Each refrigeration machine shall be mounted as specified. The condenser and compressor shall be rigidly connected on a common base before connection to the vibration isolators.

3) Installation and Commissioning

- A fully trained and authorised commissioning engineer from the refrigeration machine manufacturer shall supervise all tests including the pressure testing, evacuation, dehydration, charging, and initial start-up commissioning of the machines to the entire satisfaction of the S.O.; the commissioning engineer shall submit proper records and reports on the commissioning tests. This representative shall attend to the machines as and when required by the S.O. during the Contract period. The representative shall also instruct the Employer's operating personnel on the proper operation of the machines.

**G7.1.5 Cooling Towers**

(a) Induced Draught Cooling Towers

- (i) Cooling towers shall fit into the space and height allowed at the cooling tower enclosures/rooms and made readily accessible for maintenance.
- (ii) Adequate space shall be provided around the cooling towers to ensure that there are proper air intake passages and the performance of the cooling towers will not be affected.
- (iii) Cooling towers shall be selected to suit the weather conditions expected in Singapore. The cooling tower complete with water basin, fan, motor, tower in fill, etc shall be suitable for the required working conditions.
- (iv) Materials used for construction of the cooling towers shall be non-porous, smooth, corrosion and chemical resistant.
- (v) Mechanical induced draught cooling towers shall be supplied and fitted generally as shown in the Schematic drawings.
- (vi) Each tower shall have a cooling capacity as detailed in the Schedule of Technical Requirements.
- (vii) The cooling tower complete with basin, fan, motor, tower in fill, etc shall be suitable for the required working conditions.
- (viii) The design, construction and installation of the cooling towers shall be strictly in accordance with the manufacturer's recommendation and comply with the Code of Practice for the Control of Legionella Bacteria in Air-Conditioning Cooling Towers in Singapore published by the Ministry of the Environment.
- (ix) All parts of cooling tower shall be prefabricated and supplied with detailed erection drawings to facilitate field assembly.
- (x) Cooling towers shall be furnished by manufacturers normally supplying these type of equipment and who can show evidence of having completed at least ten installations of approximately the same types and capacities and have been in successful operation for at least three years.
- (xi) Cooling tower thermal performance shall be certified in accordance with the Cooling Tower Institute (CTI) Certification Standard STD-201 or equivalent.

- (xii) Cooling towers shall comply with the requirements of SS 530 and SS 553
- (xiii) Each cooling tower basins shall be fitted with electrode type level sensing switches for connection to a buzzer and indicator light in the Main Supervisory Control Panel and Computer-based Control System to alarm in case of low water level. Electrodes shall be properly secured in the basin to prevent swaying.
- (xiv) Cooling towers floor supports shall be mounted on steel springs with travel limit stops in series with 2 layers of 8 mm thick neoprene waffle pads as specified in the Schedule of Vibration Isolation Equipment.
- (xv) The isolators shall be galvanised by the manufacturer to prevent rusting in an exposed environment.
- (xvi) Cooling towers sump shall be suitably supported on adequately sized RSJ steel sections which shall be arranged to support the sump of each of the cooling tower.
- (xvii) Outer casings of cooling towers shall be constructed of fibreglass reinforced polyester suitable for a wind load of 240 kg/sq m.
- (xviii) The frame work of the cooling tower body shall be galvanised through hot dipping to protect the steel members against corrosion.
- (xix) Fan supports and bracing plates after the fan discharge shall be aerodynamically designed to eliminate noise generation when the fan is in operation.
- (xx) All hardware such as bolts, tension braces and other fasteners shall be of non-ferrous material (stainless steel).
- (xxi) In fill shall be of non-corrosive, non-deteriorating type and shall be laid in a manner suitable for easy replacement.
- (xxii) Distribution pipes in the centre of the tower (counter flow type) shall be of the free rotating type enabling even distribution of the water over the infill.
- (xxiii) Drift eliminators shall be in the form of baffles plates provided with the distributing pipes, offering minimum resistance to air flow with maximum removal of entrained moisture from the air and shall not exceed more than 0.002% of the circulate water rate when tested to CTI or approved equivalent.
- (xxiv) Cooling tower shall be complete with the standard connection for makeup, quick fill, bleed, drain and over flow pipes, and a permanent access ladder to provide convenient access to all parts of the tower.
- (xxv) The drain shall be located at the lowest point of the basin with a drain valve to enable easy and complete draining of the entire system.
- (xxvi) Variable speed drive for each of the fan of the cooling towers shall be provided. The cooling tower fan motors shall be compatible with the variable speed drives. The fan motor shall be totally enclosed, fan cooled and weatherproof type.

- (xxvii) Fans shall be of high efficiency, adjustable pitch axial or propeller type, fabricated from corrosion resistant materials, specially selected for cooling tower service and be complete with non-corrodible bird screens on the discharge outlet. Bearings shall be of self-aligning ball type.
- (xxviii) Vertical division plates shall be fitted at the lower level inside of the tower to keep the wind from blowing through the open sides.
- (xxix) A horizontal nonferrous fine mesh shall be fitted below the filler to prevent noise from splashing of water.

#### **G7.1.6 Water Treatment System**

##### **(a) Water Treatment**

- (i) The Contractor shall engage an approved specialist water treatment company to supply and install water treatment systems complete with all necessary metering pumps, chemical tanks, pipework, timers, automatic controllers, electrical control panels, chemicals, etc to provide suitable treatments for condenser water and chilled water systems so as to prevent corrosion, fouling, scale formation, micro biological growth, micro-organism growth (including legionella bacteria) and also to control pH value of the water within the range of 6.5 to 7.5. The Contractor shall ensure compatibility of the chemicals and automatic operation of the air-conditioning installations to enable the latter to operate at high efficiency, and without deterioration or corrosion. The water treatment systems shall comply with the Code of Practice for the control of Legionella Bacteria in Air-Conditioning Cooling Towers in Singapore, published by the Ministry of the Environment.
- (ii) The specialist water treatment company shall supply all necessary chemicals for operation of the installations during the Maintenance Period and for 12 months thereafter. The chemicals supplied shall meet the requirements for discharge of effluent to sewer and the Code of Practice for the Control of Legionella Bacteria in Air-Conditioning Cooling Towers in Singapore.
- (iii) The specialist water treatment company shall commission the water treatment system, implement the appropriate programmes to service on a monthly basis and be responsible for maintenance of the equipment during Maintenance Period. Test kits shall be supplied for analysis of the circuits, together with log sheets, product bulletins and dosing instructions. Routine test results are to be recorded in the log book and submit to the S.O., reports and recommendations for remedial actions. In addition, a formal record book with all the required information as listed in the Code of Practice for the Control of Legionella Bacteria in Air-Conditioning Cooling Towers in Singapore shall be maintained.

##### **(b) Automatic Water Treatment System for Condenser Cooling Water Circuit**

- (i) The instrument shall automatically measure and control corrosion, dissolved solids and pH value. Corrosion shall be dependent on linear polarisation for instantaneous corrosion measurement. Colorimetric or proportional methods of corrosion control are not acceptable.

- (ii) The instrument shall automatically control dosing pumps for the addition of corrosion, scale inhibitor and pH value control chemicals to the circuit and a solenoid valve to bleed the cooling water circuit.
  - (iii) The instrument shall automatically control dosing assembly for weekly slug dose additions of biocide to control micro biological growth (including legionella bacteria).
  - (iv) Chart recorders shall be provided for recording of corrosion, dissolved solids, pH value and dosing pump operations.
  - (v) The instrument shall have alarm functions which can be relayed to the Main Supervisory Control Panel and shall be suitable for connection to the Computer based Control System.
  - (vi) All dosing tanks shall be 0.2 cubic metres capacity, have drip trays and be mounted on stands. Taps shall be installed over the dosing tanks.
- (c) Chilled Water Circuits
- (i) A corrosion and scale inhibitor shall be introduced into the system via a bypass pot which shall be installed across the circulation pump. The bypass pot shall include a funnel, isolating valves, drain valve and overflow vent. The drain line shall run to sewer. Plan of the drain pipeworks connection to sewer to be submitted to the S.O. for approval.
  - (ii) The Contractor shall submit detailed technical proposals complete with catalogues/drawings of the proposed water treatment systems for the S.O.'s review and approval prior to order and installation of the systems.

#### **G7.1.7 Expansion Tank**

- (a) The Contractor shall supply & install expansion tank as per specification and Tender Drawings. Where indicated on Tender Drawings expansion tank shall be of glass-reinforced plastics (GRP) to conform to SS245.
- (b) The minimum operating water level shall be 1.2 m above highest point in the chilled water system.
- (c) Each tank shall complete with ball float valve, overflow, screen vent, drain and outlet connections.
- (d) Ball valves shall be fitted at approximately two-thirds tank level with the ball float adjusted to close valve at one-half tank level, and to reset at bottom of tank when empty.
- (e) Supply & install water level gauge complete with gauge cocks, blow-off cocks and slight glass fitted to the side of the tank.
- (f) Low-level alarm switch is to be provided to initiate an audio-visual alarm at the Central Control Panel (CCP) or Building Management System (BMS) console.
- (g) Tank support shall be of welded steel angle section construction and ensure tank is laterally braced.
- (h) The whole tank shall be treated to prevent corrosion and properly insulate against heat gain in accordance with BS 5422.



- (i) Where identified the GRP tanks shall be constructed in accordance with SS 245 with 1.0m× 1.0m nominal size panels and complete with covers, piping nozzles. Panels are to be hot press moulded to withstand local outdoor conditions.
- (j) After treated tanks internally with PVC or other suitable anti-corrosive paint. Externally shall be insulated for chilled water application exposed to weather or within plant rooms with 50 mm thick cork sheet covered with 25 mm thick waterproofed plaster supported by expanded metal beadings.
- (k) Insulated tanks located inside plant rooms shall complete with 50 mm thick, 32 kg/m<sup>3</sup> glass fibre and jacketed with 0.6mm aluminium sheet.
- (l) Provide polyurethane insulation to tanks to be compliant to Fire Code.
- (m) Exposed tanks under weather condition are to have waterproofed plaster covering supported by expanded metal beadings. Paint lightweight metal constructions liable to distortion 2 coats zinc-rich paint after proper preparation with all cut edges and the like similarly treated. Bolts, nuts and washers shall be electro-galvanized.

#### **G7.1.8 Condenser and Chilled Water pumps**

- (a) Pumps
  - (i) Condenser cooling water and chilled water pumps shall be supplied generally as shown in the Schematic Drawings.
  - (ii) The pumps shall be of the following configuration:
    - 1) Centrifugal;
    - 2) Constant speed;
    - 3) Single stage;
    - 4) End suction or;
    - 5) Horizontal split casing type; and
    - 6) Or unless otherwise specified in the Particular Specifications or Tender Drawings.
  - (iii) The pumps shall be complete with driving motors, mating flanges, air release, volute drain connections and base plates.
  - (iv) The pumpsets shall be selected for their particular applications in regard to water temperatures and conditions, suction head, static head, lift and working pressure.
  - (v) Each pumpset shall be capable of circulating not less than the nominal rate specified in the Schematic Drawings against the system resistance.
  - (vi) Contractor is required to carry out the necessary calculations relating to the pressure drop of the equipment selected by them, and of the system piping so that the pumpset is capable of providing the necessary output at the designed head.
  - (vii) Pump efficiency, when operating at the specified duty, shall be within 5 per cent (5%) of the maximum available efficiency for the particular pump.
  - (viii) Pump performance characteristic curves with the predicted operating point shown shall be submitted to the S.O. for approval.

- (ix) Nominal duties of the pump are given in the Particular Specifications and Drawings and may be used as an approximate guide only.
- (x) Pump impeller diameter shall not exceed 0.9 of the maximum impeller diameter capability of the pump housing to reduce the possibility of tonal effect.
- (xi) Pump casings shall be of close grained cast iron or cast steel suitable for the operating pressure intended and so designed to provide smooth flow with gradual changes in velocity. Impellers shall be of bronze and designed to give non overloading characteristics over large range of head variations.
- (xii) Each impeller shall be statically and dynamically balanced to ISO 1940/1, balance quality grade G6.3. Documentary proof of the balancing results shall be submitted to the S.O. for review.
- (xiii) The pump shafts shall be of machine ground stainless steel mounted in renewable bronze sleeve bearing.
- (xiv) Pump glands shall be of the mechanical seal type but stuffing boxes with bronze glands will be considered as alternatives to the base offer as specified. The design speed of the pumps shall not exceed 1,500 rpm.
- (xv) Each pump shall be directly driven by a totally enclosed fan cooled motor. Motor shall be equipped with sleeve bearings and speed shall not exceed 1,500 rpm unless otherwise approved by the S.O..
- (xvi) The Contractor shall produce manufacturer's selection charts and data to verify that coupling of the pump to the motor will meet the duty specified using standard impellers to the approval of the S.O.; otherwise, the Contractor shall provide at his own cost an indirect drive arrangement using a minimum of three V belts of matched size with a service factor of 150 %.
- (xvii) Each pump and motor combination shall be mounted on an inertia block constructed of reinforced concrete. The depth of the base shall be at least 1/12 of the longest side of the inertia block or at least 150 mm thick whichever is greater.
- (xviii) The Contractor shall provide complete welded steel channel frames with base plate for casting reinforced concrete in forming the inertia blocks. Details of the inertia blocks shall be as shown on the Standard Drawings.
- (xix) All pipe fittings, elbows, etc, before the flexible connections shall be rigidly connected to the inertia block.
- (xx) Vibration isolators for all pumps shall be as specified in Schedule of Vibration Isolation Equipment.
- (xxi) The Contractor shall submit samples of vibration isolators for approval before site construction can commence.
- (xxii) Bolt down type metal guard shall be installed over each coupling with holes neatly cut and trimmed for Tachometer readings.

### **G7.1.9 Electric Motor**

(a) General

- (i) Electric motors shall be of the totally enclosed squirrel cage induction premium high efficiency motor type Class F insulation with Class B temperature rise suitable for operation in ambient temperature up to 40 °C.
- (ii) Under any conditions of operation when the fan is running at its designed speed and with the impeller blades set at the predetermined pitch angle it shall not be possible for the fan shaft power - to exceed the rated power of the motor.
- (iii) The fan impeller shall be overhung on the shaft of the electric motor and the motor shall be designed for this application.
- (iv) The motor shall be air-stream rated for continuous maximum duty.
- (v) Motors shall be selected for quiet running characteristics and efficiency.
- (vi) The power and efficiency factor for all motors over 12 kW shall be submitted with the tender.
- (vii) All motors shall be of the squirrel cage induction type specifically wound and suitable for the power supply nominated in the Tender Specification.
- (viii) All motors intended for use shall be tropic-proofed.
- (ix) Motors shall be capable of running continuously with a 5 percent drop in rated phase to phase voltage at 15 percent increase in design power.
- (x) Motor of 0.75 kW and over shall be fitted with integral positive temperature coefficient thermistors selected to afford Class 'I' Protection Motors below 0.75 kW shall be fitted with inherent overheat protection.
- (xi) Motor speed shall not exceed 1500 rpm.
- (xii) The motor shall be mounted concentrically in casing and rigidly mounted from the casing.

(b) Motor Bearings

- (i) Motors shall be fitted with ball and/or roller bearings suitable for horizontal or vertical mounting of the motor shaft. Adequate provision shall be made to carry the axial thrust of the fan impeller and motor rotor.
- (ii) Bearing must be selected to ensure that the specified noise levels can be achieved when the fan is in operation.
- (iii) The bearings shall be grease lubricated. Regular lubrication shall be either by re-packing the bearings or by introducing grease through a grease nipple or screw down greasing point located outside the fan casing or easily accessible through the access panels. The bearings shall be packed with grease in the quantity and grade recommended by the bearing manufacturer and shall be the type of grease stated on the fan nameplate. Bearings shall be effectively sealed against the ingress of dust and moisture and against leakage of lubricant other than excess grease.

**G7.1.10 Noise and Vibration Control**

(a) General

- (i) This covers the supply; delivery, installation and testing of Noise and Vibration Control equipment to be used in the isolation of the various Mechanical equipment called for in this specification.
- (ii) It is the intent of this specification that noise levels due to mechanical equipment and related services will be controlled to the design objectives stated herein, in all occupied areas. The requirements specified are considered to be the minimum precautions necessary to achieve these objectives. The entire installation shall operate without objectionable noise and vibration as determined by the S.O..
- (iii) This specification covers the supply, delivery to site, installation, commissioning and guarantee for one year of the noise and vibration control equipment as described in the specification.
- (iv) The Contractor shall select and provide all mechanical equipment to this specification, complying in particular with the requirements concerning balancing of the equipment, mounting of the equipment, alignment of driving and driven units and operating speeds.
- (v) The Contractor shall install all mechanical plant and services in accordance with the methods of installation and precautions stated herein, and such additional precautions as may be necessary to ensure that the operation of the plant does not result in noise levels or vibration amplitudes beyond the specified limits.
- (vi) The Contractor shall examine all drawings and specifications including structural sets of working documents, before commencing any work on the project and shall immediately bring to the S.O.'s attention any characteristics or properties of the building or any other factors which, in his opinion, would jeopardise or nullify the attainment of the design objectives.
- (vii) The Contractor shall guarantee that the complete plant and installation when operated within the design criteria shall acoustically perform to the noise criteria ratings specified in this Section.
- (viii) The Contractor shall be responsible for any corrective action, which may be necessary either during construction or after completion of the Works, to achieve all design objectives.

(b) Design Objectives

- (i) Octave band sound pressure levels in the various areas within the building and at certain positions outside the building due to the operation of the equipment included in this Contract shall not exceed the noise level criteria set out in the following schedule.
- (ii) Where dispute arises over the classification of any area under the following schedule, the S.O.'s determination-of the space type and function of the area, as listed below, shall be final.

| Area                                                   | dBA |
|--------------------------------------------------------|-----|
| Cinemas, theatres.                                     | 25  |
| Private executive type office                          | 35  |
| General Offices, other private or semi-private offices | 45  |
| Conference rooms                                       | 35  |
| Air-conditioned classrooms                             | 35  |
| Hotel bedrooms                                         | 35  |
| Places of public resort e.g. shops                     | 45  |
| Circulation areas e.g. staircases, lobbies, car parks  | 45  |

The specified interior noise level limits apply to every position from 1 m to 2 m from the floor not closer than 2 m from any air outlet, or equipment.

- (iii) Vibration levels criteria shall be subjectively “virtually imperceptible” with respect to feelable vibration, for the majority of people, and shall at the same time, not allow re-radiated structure borne noise levels to exceed approx. NC 35 to minimise contribution to airborne and in the audible region:
    - 1) Minus 60 dB re IG from 2 Hz to 10 Hz; and
    - 2) Minus 55 dB re IG from 10 Hz to 31.8 Hz.
  - (iv) Plantrooms are excluded from noise level restrictions, provided the operation of equipment does not raise the noise level above the acceptable limits in any area subjected to the above noise level restrictions.
  - (v) In general, the control of airborne noise transmission through building elements of noise generated within the plant areas by equipment, shall also be the responsibility of the Contractor. The Contractor shall be responsible for and ensure that all penetrations in walls, floors and ceilings provided for pipes, ducts, conduits, etc. are isolated and sealed around the penetration. In all cases, masonry and other structure surrounding the plant shall not be in direct contact with equipment.
- (c) Anti-Vibration Equipment/Techniques
- (i) The following summarises the type and nature of the noise and vibration control techniques and equipment to be used in the isolation of the various mechanical equipment as described in this specification:
    - 1) Steel Spring in Series with Neoprene Pads
      - Unless specified otherwise, spring type isolator shall be unhoused, free-standing laterally stable with levelling bolts rigidly bolted to inertia blocks/base-frames. In general, diameter shall be comparable with compressed height and shall not be less than 0.8 of compressed height at rated load, and horizontal spring stiffness shall be approximately equal to vertical stiffness.

- Spring shall have a minimum additional travel to solid equal to 50 % of rated deflection, and in this condition, shall not exceed elastic limit of spring.
- All springs shall be mounted with adequate clearance from brackets and shall be clearly in view for critical inspection.
- Where restrained mounts are specified, housing shall be used that includes vertical limit stops. A minimum clearance of 10mm shall be maintained around the restraining bolts and between the housing and spring so as not to interfere with spring action. Limit stops shall be out of contact during normal operation. Mounts used outdoors shall be hot dipped galvanised.
- All steel springs shall be installed in series with two layers of neoprene ribbed or waffle pattern pads (min. 8mm thickness) as described below.
- The springs shall be selected to support the calculated loads with the minimum spring deflections as specified without being compressed solid, or bridged out, or exceeding the manufacturer's maximum rated loads and rated deflections under any operating condition. In the selection of the springs, the Contractor shall also take into account the weight distribution of the equipment under normal operating conditions, weight of unsupported pipes/ducts, and any large dynamic forces due to fluid movement, torque reaction and starting/stopping. The Contractor shall show, on the shop drawings submitted, the make and model number of the isolators selected and their individual calculated loads.

(d) Inertia Blocks

- (i) The mechanical equipment shall be bolted directly on to the integral concrete inertia blocks. All concrete referred to in this specification should have a density of at least 2240-2400 kg/cu.m. (140-150 lbs/cu.ft.).
- (ii) In general, the length and width of the inertia block should be at least 50 % greater than the length and width of the supported equipment, except where otherwise specified.
- (iii) The weight of the inertia block, generally determined by thickness limits, and inertia block/equipment mass ratios shall be as specified.
- (iv) The base shall consist of a concrete slab cast into a welded steel base-frame assembly. Frames shall be welded steel channels, of min. 150 mm (6 ") thickness or the same thickness as the Inertia Block and shall be reinforced with welded-in 12 mm (1/2 ") steel reinforcing rods or angles at 150 mm (6 ") centres each way. Anchor bolts shall be fixed into position, and housed in steel bolt sleeves to allow minor bolt location adjustments or alternatively, pockets shall be cast into inertia block to permit the later insertion of anchor bolts.
- (v) A suitable base former of min. 10 mm plywood or equivalent shall be included for forming the slab. Steel channel isolator brackets shall be welded directly to channel frames, and suitably located to accommodate the height of the deflected springs and inertia block clearance with plinth.

- (vi) There shall be adequate clearance (min. 20 mm) all round the springs to assure that there is no contact between any spring and any part of the mounted assembly for any possible alignment or position of the installed inertia block. The clearance between inertia block and floor/plinth shall be at least 75 mm (3 ").
- (vii) The Contractor shall supply and install all welded structural steel and reinforcement required for the inertia blocks. All installation work concerning these items is included with this Contract.
- (e) Ribbed or Waffle Pattern Neoprene Pads
  - (i) To be used where specified - 8 mm (5/16") - 9.5 mm (3/8 ") thick 40 durometer pads as normally supplied by approved manufacturers.
  - (ii) For typical applications, 40 durometer type neoprene pads should be loaded to a surface weight of approximately 35,000 kg/sq.m. (50 lb/sq.ft.).
  - (iii) Where two or more layers of neoprene pads are used, individual layers shall be separated by 20 g thick steel shims.
- (f) Resilient Hangers
  - (i) Resilient hangers are to be used for resilient suspension of pipes and ducts, where specified.
  - (ii) For specified deflections in excess of 12mm, use spring in series with neoprene type and for deflections less than 12mm use neoprene in shear type hangers.
  - (iii) During installation, inspection shall be made to ensure that no hangers are compressed solid or bridged out, and that the stipulated minimum deflections are achieved.
- (g) Neoprene-in-Shear Isolators
  - (i) Neoprene-in-shear isolators shall be of the double deflection type.
  - (ii) The isolators shall incorporate cast-in tapped steel load plates to permit direct bolting to supported equipment. Cast-in drilled base plates shall also be incorporated for direct mounting to plinths.
  - (iii) The isolators shall be selected to achieve the minimum deflections specified, without exceeding the Manufacturer's published maximum permissible loading.
- (h) Refrigeration Plant
  - (i) Centrifugal and reciprocating chillers shall be mounted on steel springs in series with neoprene. Limit stops are required to limit travel when machine is drained. A steel base-frame suitable for point loading of the chillers shall be provided as approved by the S.O..
- (i) Miscellaneous Equipment
  - (i) All equipment located in plantrooms above or under occupied areas and capable of producing noise or vibration shall be isolated from the structure. Acoustical isolation by neoprene/fibreglass cork pads is required for tanks, calorifiers, electrical switch-boards, etc.

#### **G7.1.11 Pipework, Valves and Fittings**

(a) Pipework

- (i) This Section of the Specification contains requirements necessary to design, provide, and install, complete, the pipework and fittings.
- (ii) This specification should be read in conjunction with the relevant drawings schematics. An indicative location of the pipework is shown on the general arrangement proposed for the site on the drawings.
- (iii) All pipeworks delivered to site shall be new and colour banded to identify the different grades.
- (iv) Pipeworks shall be installed with correct falls, slopes or gradient to ensure proper venting and draining.
- (v) Pipeworks installed in trenches, ducts, voids and inaccessible areas shall have welded joints except where screwed or flanged joints are necessary for connection to valves, etc. and where welding is not allowed or possible.
- (vi) No pipe joints shall be permitted within the thickness of walls, floors and beams, etc.
- (vii) For sprinkler protected area, the pipe supports and pipe penetrations shall be protected by sprinkler system.
- (viii) Except only for non-combustible materials or services, all others are not allowed to be placed within 1m before and after the penetration

(b) Pipe Materials

- (i) All chilled water supply & return pipes shall be Class 'C' black steel to BS EN 10255 and BS EN 10216.
- (ii) All condensate drain pipes shall be the unplasticised PVC pipe complying with Singapore Standard SS 141.

(c) Pipe Fittings

- (i) Galvanised pipe fittings shall be used on galvanised pipe work and black pipe fittings on black steel pipe work.
- (ii) In general, all fittings shall be of the malleable iron type. However, they should be entirely suited to the working and test pressures of the circuit concerned.
- (iii) All pipes 50 mm diameter and smaller may be screwed using litharge and glycerine as a sealing compound. All pipe lengths to be cut with standard pipe cutters of jig fitted pipe saws, threads cut with standard thread dies, and ends reamed before assembly. All screwed fittings up to diameter 50 mm shall conform to BS EN 10242.
- (iv) All pipes and joints 65 mm diameter and above shall be connected by either welding or welded flanged except at fittings and valves where flanged connections shall be used. Welded fittings shall comply with SS EN 10224 or BS EN 10253-1.
- (v) Welded flanges shall be of the "slip-on" type jacketed with approved gasket material and sealed using litharge and glycerine and bolted. Welded flanges shall comply with EN 1092-1 and shall be for the service pressure encountered.



- (vi) Connections of 25 mm diameter and smaller to 50 mm diameter and larger shall be butt welding to the wall of the pipe. No drilling and tapping on the wall of the large pipe will be allowed.
  - (vii) Pipework connection of 25 mm to 50 mm diameter shall be by reducing tees and bushings where necessary.
  - (viii) Standard reducing pieces shall be used throughout the whole installation.
  - (ix) All equipment and valve connections shall be arranged with unions joint up to and including 50 mm diameter pipe, and flanges joint over 50 mm diameter pipe for ease of disconnection.
- (d) Pipe Joint
- (i) All pipe jointing shall be carried out in a proper workman like manner.
  - (ii) Steel Piping
    - 1) Welded
      - Joints in steel pipes made by welding shall be to the approval of the S.O.;
      - All welded joints shall comply with BS EN 10253-1. Pipe ends shall be prepared by machining, grinding, or machine gas cutting or hand flame cutting with subsequent grinding;
      - For pipes which are intended to be in axial alignment, the plane of the ends shall be square with the axis of the pipe;
      - All welding shall be done by approved competent welders and such certification are to be submitted to S.O.;
      - Where welds fail to comply with the requirements of the S.O., the faulty welds shall be cut out and rectified; and
      - Where necessary protection from weather shall be provided by the Contractor to enable welding operation to be performed properly.
    - 2) Flanged
      - Where welded joints are impractical or where required for erection purposes or at connections to fittings on steel pipes 50 mm or over or at all flanged valves, flanges shall be fitted and welded on to pipes to approval;
      - Jointing flanges shall be truly parallel to each other so that bolts are used only to make joints tight and not to pull up flanges not true alignment. All bolts and nuts shall be of bright steel of approved manufacture. Flanges shall conform to EN 1092-1; and
      - Taylor metallic corrugated jointing rings with approved jointing paints or other approved material shall be used between flanges.
    - 3) Unions
      - Where welded joints are impractical or at connections to fittings on pipes under 50 mm, approved steel socket steam union may be used.

(e) Pipe Bends and Tees

- (i) All bends and tee fittings shall conform to BS EN 1254-1 & 2 and shall be of approved manufacture. All to be dezincification for diameter of sizes 6 mm to 67 mm.
- (ii) Preformed bends shall have a radius of not less than twice nominal pipe size for short radius, and four times nominal size for long radius requirements.
- (iii) Tees for steel pipes 50 mm and over shall be slip-on or other approved bevelled end. Pipes less than 50 mm diameter tees may be welded.

(f) Piping Supports

- (i) All piping shall be supported by approved hangers and supports placed at not more than the following intervals:

|                                  |                | Support Spacing      |                    |
|----------------------------------|----------------|----------------------|--------------------|
| Pipe Material                    | Pipe Size (mm) | Horizontal pipes (m) | Vertical pipes (m) |
| Steel Pipe                       | 15             | 1.2                  | 1.2                |
|                                  | 20             | 1.8                  | 1.8                |
|                                  | 25             | 2.1                  | 2.1                |
|                                  | 35             | 2.4                  | 2.4                |
|                                  | 40             | 2.7                  | 2.7                |
|                                  | 50             | 3.0                  | 3.0                |
|                                  | 65             | 3.3                  | 3.3                |
|                                  | 80             | 3.6                  | 3.6                |
|                                  | 100 – 250      | 4.1                  | 4.1                |
|                                  | 300 – 400      | 5.5                  | 5.5                |
| PVC and Other Thermoplastic Pipe | 15 – 65        | 1.2                  | 1.2                |
|                                  | 80 – 150       | 2.0                  | 2.0                |

- (ii) Hangers for horizontal pipe runs shall be split ring adjustable type or other approved design hung on round steel rods. Brackets or clamps may be used where pipelines run along walls, columns or ceilings.
- (iii) A hanger shall be placed not more than 610 mm each change of direction of pipes, preferably on the side with the longest run.
- (iv) Hangers for horizontal pipe runs shall allow for expansion of pipelines and shall have provision made for adjusting grade and alignment.
- (v) Hanger rods of the following minimum sizes or equivalent shall be provided:

| Pipe Size (DN) | Rod Diameter (mm) |
|----------------|-------------------|
| 15 to 50       | 10                |
| 65 to 80       | 12                |
| 100            | 16                |
| 150            | 20                |
| 200 to 600     | 2 x 20            |

- (vi) Where supports are to be fastened around insulation, heavy density insulation or approved type saddles shall be placed between pipe work and support and moulded into adjacent insulation in an approved manner so that there will be no future deterioration of the support and no reduction of the effective insulation of the pipeline.
  - (vii) Straight pipe runs in any direction in excess of 30 mm on one side of an anchor point shall have axial expansion bellows of an approved design.
  - (viii) All vertical risers shall be parallel to walls and column lines and shall be straight and plumb. Where possible, vertical runs of pipes shall be grouped on pipe racks. Racks shall consist of shaped steel frames fixed to walls and ceilings. The pipes shall be carried in saddles, bolted or screwed to the racks.
  - (ix) All buried pipes shall be laid in accordance with manufacturer's recommendations or local codes and details and design are to be submitted to S.O..
  - (x) Pipe hangers or brackets shall be properly isolated to prevent vibration and noise transmission to the building and special care shall be taken in making piping connections to pumps and all other vibrating equipment.
  - (xi) The first three hangers/supports from all Mechanical equipment shall be spring in series with neoprene types.
  - (xii) All other supports/hangers unless otherwise specified, shall incorporate an isolation medium of 8 mm thickness such as waffle pattern neoprene, felt or high density fibreglass between pipe and hanger/support or between hanger/support and wall. This applies to all piping installed on this project, whether in horizontal runs or vertical pipe risers.
  - (xiii) All pipe penetrates through wall, floors and ceiling shall be isolated from direct contact with the structure, with a fibreglass isolated sleeve caulked with a non-setting resilient compound on both sides of the penetration.
- (g) Valves
- (i) All valves shall be manufactured to BS EN ISO 9001, or EN 29000 series, product-tested to relevant British Standard. It shall be placed in accessible position for operation and maintenance. The size of valves shall be of the same diameter as the pipe to which it is connected.

- (ii) All valves supply shall be of an approved manufacture and have a working pressure of not less than 1050 kPa and compatible to system's working pressure.
  - (iii) All gate and globe valves 65 mm and above shall be flanged cast iron body with internal spindle to BS ISO 7121.
  - (iv) All check valves 50 mm and smaller shall be of screwed bronze body type to BS EN 12164 and BS EN 12165.
  - (v) All check valves 65 mm and above shall be of the dual-check type consisting of a cast iron body and two bronze half-plates hinged on the valve stem to BS EN 13959. The seat shall be BUNA-N and spring shall be of stainless steel.
  - (vi) All butterfly valves shall have a bubble-tight shut-off rating of not less than 1050 kPa and compatible to system's working pressure. They shall be of canter line or keystones and listed by Underwriter Laboratories (UL).
  - (vii) All butterfly valves shall be of flanged cast iron body with bronze discs and BUNA-A rubber seats. The valve stem shall be of stainless steel with either 'O' ring type seals or non asbestos material packed glands.
  - (viii) A lever actuator shall be used for the operation of all manual control valve. The actuator shall be smooth in operation and shall require minimum force to operate. All valves with diameter larger than 100 mm dia., gear operated handlers must be used.
  - (ix) All automatic flow control valves shall conform to BS 7350. Do not fit valves before the piping system has been tested, flushed and cleaned.
  - (x) All fixed orifice double regulating valves up to 50 mm diameter shall conform to BS 7350 directly coupled to a flow measurement device comprising male/female threaded nipple type carrier with integral orifice ring and two double seal pressure test points. Cast-iron double regulating valves of 65 mm and above is to comply with BS 7350, flanged PN16. Single piece stainless steel square edged flow measurement orifice plate and carrier, with two double seal pressure test valves, to fit between the valve outlet flange and mating flange.
- (h) Strainers
- (i) All strainers installed in pipeline shall be of 'Y' type with easily removable stainless steel strainer cartridge. Strainers shall be arranged with isolating valves so that they may be cleared without draining the system.
  - (ii) All strainers 50 mm and smaller shall be of screwed bronze body type.
  - (iii) All strainers 65 mm and above shall be of flanged cast iron body type.
- (i) Flexible Connections
- (i) All piping connections shall be soft spherical single convolution EDPM rubber joint flexible connectors of high vibration isolation efficiency, resilience and noise absorbing properties, corrugated stainless steel connections of approved manufacture. They shall have integral flanges and be bolted on to the pipelines using flanges for 50 mm diameter and larger. For smaller than 50 mm, screwed type wire and fabric reinforced moulded high pressure connectors shall be used.

- (j) Safety & Air Relief Valves
- (i) Automatic relief valve shall be installed at all high sections of pipe works where air may collect. The discharge from these vents shall be piped via copper pipes of appropriate size to the nearest waste drain pipe or trap.
- (k) Constant Flow Valve
- (i) Flow control valves shall be factory calibrated, direct acting, automatic pressure compensating type. Each valve shall be limited to within +5 L/s accuracy regardless of system pressure fluctuations. Valve control mechanism shall consist of a tamper proof, stainless steel cartridge assembly with open chambers and unobstructed flow passages. Cartridge assembly shall include a self-cleaning, spring-loaded moving cup guided at two separate points and shall utilise the full available differential pressure to actuate without hysteresis or binding. Four differential pressure ranges shall be available with the minimum range requiring less than 2 psi. Each valve to be provided with a metal tag, chain and stamped for system identification. Pressure taps and quick disconnect valves shall be provided with ferrous bodies. All hydronic system flow control valves shall be provided by one Manufacturer.
- (l) Instruments
- (i) All Instrumentation must comply with SSCP 591 - Long term measurement of central chilled water system energy efficiency.
- (ii) All pressure gauges shall be of 100 mm dial 'Bourdon' tube type to BS EN 837-1, except those in plantroom to be 150 mm diameter, with enamelled mild steel case with chrome bezel, substantial glass face, and phosphor bronze 'Bourdon' tube. Dial face to be white with black scale graduations and numbering, with the range selected so that normal operation is near to the middle of the scale.
- (iii) The gauge shall have an adjustable reference pointer. Approved syphon connection, gauge cocks and vibration snubber shall be provided.
- (iv) Gauges overall accuracy to be one per cent (1 %) of scale range and comply with the table below:
- |                        |      |      |     |     |     |
|------------------------|------|------|-----|-----|-----|
| Pressure (bar)         | 10   | 4.5  | 3.0 | 2.5 | 1.5 |
| Gauge (bar)            | 0-20 | 0-10 | 0-6 | 0-5 | 0-3 |
| Figure Intervals (bar) | 2    | 1    | 1   | 1   | 1   |
| Intermediate (mbar)    | 500  | 100  | 100 | 100 | 100 |
- (v) Fixed thermometers shall conform to BS EN 13190, 150 mm diameter dial size in main boiler / water heater and other plantrooms, and 100 mm diameter dial size elsewhere, with black enamelled mild steel case with chrome bezel and substantial glass face. Stem immersion length normally 100 mm with bottom or back connection.

- (vi) Thermometers shall match the pressure gauges, align wherever possible, and shall complete with stainless steel or brass well with screwed cap and 50 mm lagging extension. Fill wells with thermal conducting agent and keep covered until thermometers are installed.
- (vii) Install thermometers vertically in the following locations:
  - 1) Chiller evaporator and condenser inlets and outlets;
  - 2) Cooling tower inlets and outlets;
  - 3) Water/water heat exchanger inlets and outlets; and
  - 4) On air handling unit coil connections.
- (viii) Install wells for test thermometers at water inlet and outlet to each evaporator of the water chilling units as specified.
- (ix) Thermometers shall be located for easy read from the usual walking space at floor level or on platforms.
- (x) Select the thermometer range appropriate to the particular service.
- (m) Pre-Insulated Chilled Water Pipe Insulation
  - (i) All chilled water pipe work and condensate drain pipe are to be insulated, Pipe insulation used must be fire retardant and shall comply with SS 553.
  - (ii) Pipe shall be insulated with machine injected CFC-free polyurethane foam 48 kg/m<sup>3</sup> density and compressive strength 260 kPa minimum.
  - (iii) Thermal conductivity: 0.023 W/mK at 10 °C.
  - (iv) Minimum thickness of insulation shall be as tabulated below for phenolic foam preformed insulation to BS EN 13166, Type A, free of water-soluble chlorides, with factory-applied bore coating.
  - (v) Phenolic Foam Physical properties:
 

|                            |                                             |
|----------------------------|---------------------------------------------|
| Maximum System Temperature | 120 °C                                      |
| λ value                    | 0.020 W/mK (mean temperature 10 °C)         |
| Density                    | 35 kg/m <sup>3</sup> minimum                |
| Surface spread of flame    | Class 1, BS 476-7                           |
| Ignitability               | Class P, BS 476-12                          |
| Fire propagation           | 'I' less than 12, 'I' less than 6, BS 476-6 |
| Smoke obscuration          | Less than 5 %, BS EN ISO 5659-2             |
  - (vi) Pipe shall be jacket with 0.6 mm minimum thickness spirally-wound galvanized mild steel strip.
  - (vii) The insulation thickness offered shall follow closely to manufacturer's recommendation.
  - (viii) Generally, the insulation must be effective to prevent condensation on insulated pipe surface in all locations where the pipe is insulated, with various working conditions.

- (n) Condensate Water Drain Pipe Insulation
- (i) All condensate drain pipe insulation shall be of expanded nitrile rubber elastomeric insulating material which achieves Class 'O' rating and incorporates a smooth external impermeable barrier surface.
  - (ii) Manufacturers' confirmation of the maximum system temperature allowable is to be obtained.
  - (iii) Flexible Closed Cell Insulation Physical properties:
 

|                                 |                                              |
|---------------------------------|----------------------------------------------|
| Maximum System Temperature      | 150 °C                                       |
| value (mean temperature)        | 0.037 W/mK (20 °C) 0.039 W/mK (40 °C)        |
| Density                         | 90 kg/m <sup>3</sup> minimum                 |
| Surface spread of flame         | Class "O", BS 476-7                          |
| Fire propagation                | 'I' less than 12, 'I' less than 6 (BS 476-6) |
| Water vapour permeance          | 0.01 g/s MN                                  |
| Water absorption (by volume)    | 2.5 % (28 days)                              |
| Ozone resistance                | (ASTM-D-1171) 'no cracking'                  |
| Resistance to oils and greases: | 'high'                                       |
- (o) Testing
- (i) All pipe work shall be tested in convenient sections as the work proceeds. The S.O. shall be notified when testing is to be carried out so that he may have the opportunity of witnessing the test should he so desire.
    - 1) Water Piping
      - As water pipe reticulation proceeds and before pipes are concealed or insulated, hydrostatic pressure tests shall be applied and held for twenty-four hours;
      - The piping test pressure to be applied shall be 14 kg/cm<sup>2</sup> (200 psi) or 2 times the system working pressure whichever is greater. In the event of any drop of pressure during the test periods, the leaks shall be found, make good and the line re-tested. Pressure shall not show a drop of more than 2 % in twenty-four hours; and
      - Particular care must be taken to exclude all air from the system during tests.
- (p) Painting and Labelling
- (i) General
    - 1) Unless otherwise specified, all exposed surfaces including trunking and cable tray, ductwork, equipment etc, shall be cleaned and painted.
    - 2) All ferrous metal surfaces without protective finishes shall be painted, except surfaces of moving parts which shall be thoroughly oiled and greased as required.

- 3) Non-ferrous surfaces may be left unpainted unless called for in this specification or required by the S.O. for the purposes of colour coding and identification.
- 4) All bare surfaces requiring painting shall first be given a priming coat followed by an undercoat and two finishing coats.

(ii) Colour Scheme

- 1) All services shall be painted in accordance with the following colour scheme:

| Services                                             | Colour               |
|------------------------------------------------------|----------------------|
| Drain                                                | Black                |
| Hangers, supports, framework, valves, fittings, etc. | To match surrounding |
| Registers, diffusers, grilles                        | To match surrounding |
| Surface behind registers, diffusers and grilles      | Matt black           |
| Air handling units, fan coil units                   | Jade Green           |
| Ductwork                                             | To match surrounding |

- 2) All major items of equipment shall be identified with approved names and/or numbers of suitable size in proportion to the size of the respective items.
- 3) Pipelines shall be painted in contrasting colour directional arrows adjacent connections, valves and branches and at intervals of not more than 2.5 mm. These arrows shall be 75 mm long on pipes up to 50 mm diameter and 150 mm long on pipes over 50 mm diameter.
- 4) All stop valves and control valves the function of which are not obvious shall be provided with a non-corroding metal or laminated plastic identification tag. The tag shall indicate the service, and the area or items which the valve serves

**G7.1.12 Air Handling Units**

(a) General

- (i) The air handling units shall be of modular double skin construction with multiple sections like mixing box with dampers, filter section, cooling coil section and fan section.
- (ii) All the units specified under this contract shall be horizontal draw through type with fan discharge configuration matching the site requirements.
- (iii) The overall dimensions and configuration of sections shall be compatible with the existing site arrangements for equipment plinths, ducting, piping etc. as a whole.



- (iv) All the modular sections shall be factory fabricated and field fabrication of any section including mixing box section shall not be allowed.
  - (v) Lifting Provisions: Lifting provisions shall be included on air handling units as an integral part of the equipment structure. The Contractor shall assess each unit's requirements for maintenance based on equipment sizes and weight and make an assessment of lifting provisions. The Contractor shall submit assessment of requirements for lifting provisions for S.O.' review and approval prior to procurement of the equipment.
  - (vi) Where AHUs have lifting provisions required for maintenance, they shall be provided with a monorail lifting beam designed to be integral with the unit casing structure that has a safe working load rating equal to or greater than the largest equipment load within the AHU, e.g., fan motors.
  - (vii) Anti-corrosion tests for all outdoor units shall be conducted by the equipment manufacturer and compliant to accelerated test of ASTM B117 of 3000 hours salt spray test and DIN 50018-2013 of 15 cycles moist Sulphur dioxide test.
  - (viii) The outdoor units shall include factory applied heavy anti-corrosion treatment which includes coating to the heat exchanger, bottom frame assembly, mounting plate assembly and partition plate assembly. The treatment shall come with a base coat and top coat of  $\geq 50\mu$ . Coating material for the outdoor unit shall include a primer base coat and top coat of polyurethane or enamel. Units shall come with SUS410 self stainless steel screws, bolts or nuts.
- (b) AHU Submittals
- (i) Unit identification as referenced in Contract Documents.
  - (ii) Manufacturer's name and model number.
  - (iii) Descriptive specifications, literature, and drawings.
  - (iv) Dimensioned layout, including dimensioned curb layout.
  - (v) Dimensions and weights for unit, including fully assembled and shipping sections.
  - (vi) Acoustics:
    - 1) Fan sound power level data (ref. 10 to power minus 12 Watts) at design operating point to achieve specified sound performance levels.
  - (vii) Fans:
    - 1) Type, size, quantity, class, drive arrangement, discharge, rotation and bearings.
    - 2) Wheel type, diameter, rpm, and tip speed.
    - 3) Certified fan performance curves indicating:
      - Relationship of flow rate to static pressure (total internal and external) for 100%, 75%, 50%, and 25% fan speeds (internal, external, and total pressure data to be reported).

- Static pressure drops across internal air handler components and each air handler section (internal static pressure drop profile through air handler at the design air flow rate).
  - Brake horsepower curves.
  - Acceptable selection range (surge curves, maximum safe operating rpm, etc).
  - Static pressure, capacity, horsepower demand and overall efficiency required at the duty point, including drive losses.
- (viii) An assembly and disassembly scaled drawing that shall demonstrate that the AHU major components – fans and motors – can be safely disassembled and lifted in and out of the AHU access door or panel via the AHU manufacturer provided monorail lifting beam within the fan and motor section.
- (ix) Coils:
- 1) Type, quantity, dimensions, material of construction, coatings, if applicable, energy transfer capacity, air pressure drop, air inlet, and discharge temperature at design conditions.
  - 2) Hydronic Coils: Water pressure drop at design conditions, water discharge temperature at design conditions, piping connection data, coil fin spacing, coil row depth.
  - 3) DX Coils: Refrigerant saturated suction temperature at design conditions, refrigerant piping configuration (row split, face split, intertwined), coil fin spacing, coil row depth.
  - 4) Drain pan details.
  - 5) Coil pull details and dimensions for service, and maintenance. The Contractor shall coordinate the system details and installation to allow for safe access and ease of maintenance.
  - 6) Motor(s) type, quantity, and performance data, comply with General Specifications.
- (x) Air filter(s) type, quantity, and performance data.
- (xi) Unit capacities and ratings, including airflow and static pressure summary.
- (xii) Construction materials.
- (xiii) Power and control wiring diagrams, including terminals and numbers.
- (xiv) Vibration Isolation:
- 1) Vibration isolation methods with maximum deflection data.
- (xv) Additional requirements (including by others) to achieve specified vibration isolation levels.

(c) Fan & Motor

- (i) The fans shall be certified as per AMCA 210 and AMCA 300 Standards or equivalent approved. The fans shall be double width, double inlet multi blade centrifugal either back ward inclined or aerofoil type blades.
- (ii) The entire fan section shall be dynamically and statically balanced at the factory as a complete fan / motor and drive assembly conforming to ISO standard and shall be mounted on a common base framework resiliently isolated from the unit by rubber in shear or spring type vibration isolation.
- (iii) The fan blades shall be treated and coated heavy gauge steel. The fans shall be equipped with L50 life bearings of 200,000 hours.
- (iv) The fan shaft design shall not exceed critical speed at any catalogued rpm.
- (v) The drive package shall comprise of variable pitch pulley for at least  $\pm 20\%$  variation in fan rpm. Each fan drive shall include multiple v- belts.
- (vi) Motors shall be of TEFC squirrel cage induction premium high efficiency motor type Class F insulation with Class B temperature rise and suitable for operation at the local power source characteristics. The motor mounting shall allow movement on three dimensions for ease of alignment and belt tenoning.

(d) Cooling Coils

- (i) The coil section shall be of the cartridge-type, removable from the side of the casing and supported over the entire length of the coil.
- (ii) Chilled water coil shall be plate fin type with aluminium ripple corrugated fins and staggered clearable tubes with not more than five (5) fins per centimetre.
- (iii) The performance of the cooling coils shall be rated as per ARI Standard 410 or approve equivalent.
- (iv) The cooling coil selection shall ensure that the coil face velocity shall not exceed 2.5m/s at the rated airflow. Fins shall have collars, belled and firmly bonded to the tubes by having the tubes mechanically expanded into the fins.
- (v) The coil tubing shall be copper and shall have outside diameter of 13mm. Coils shall be rated to ARC Standard. Coils shall have automatic air vents, the vent outlets being piped to the drain pan with a copper pipe.
- (vi) The condensate drain pan shall be dual pitched sloping drain and shall cover the entire width of the cooling coil.
- (vii) The drain pan shall be galvanised steel, minimum 1 mm thick and coated with adequately corrosion protected with a durable mastic compound.
  - 1) Location: Under each cooling coil.
  - 2) IAQ style drain pans, complying with requirements of ASHRAE 62.
  - 3) Triple pitched for complete drainage, with no standing water in unit.

(e) Cabinet Construction

- (i) The casing design shall be based on Penta-post perimeter with double wall panels.

- (ii) The casing strength shall be designed to meet standards like European Standard EN 1886: latest edition, Casing Strength class 2A or approved equivalent.
- (iii) The removal any panel shall not affect the structural integrity of the unit. The casing shall have extruded aluminium perimeter frame.
- (iv) An in-situ CFC – Polyurethane insulation shall be applied to all the panels. The insulation shall be of adequate thickness to avoid condensation on the unit while the unit is operated but in any case not less than 50mm thick.
- (v) Units are to be weathertight and rated for outdoor use. The entire exterior of the unit shall be adequately weather protected with multiple surface finishes like baked polyester powder coating. The paint shall be ultra violet resistant, weather resistant and shall not be affected by detergent cleaning.
- (vi) Every modular section shall be accessible with proprietary access panels with leak-proof proper rubber sealing and hinged for easy opening or removal.
- (vii) A thermal break barrier shall be provided between the inner and outer wall and condensation in any part of the air handling unit shall not be accepted.
- (viii) Provide cross broken and sloped roofcap system.
- (ix) Drip shield/overhang provided on all four sides of unit.

1) Access Doors:

- Construction and thickness shall match the rest of the unit casing. Corners shall be welded for rigidity.
- Hinged, latched, and gasketed to provide weatherproof seal.
- Access doors shall be provided in sufficient number and size to allow for safe and easy removal of equipment from within the AHU.
- Doors shall open against positive pressure.
- Construction and thickness shall match the rest of the unit casing. Corners shall be welded for rigidity.

2) Intake hoods:

- Fresh air and exhaust air hoods shall be provided complete with 13mm x 13mm birdscreen and finished to match the unit. A rain gutter shall be provided on all edges of the hood.

3) Test Ports:

- Provide 25mm diameter test ports for unit air stream testing in each plenum section between each component within the AHU. Test ports shall have a tube that extends between the inside and outside of the unit and a screwed cap on the exterior to allow access. The test ports shall have been flanged on the exterior to allow air seal and shall be flanged on the interior to cover the penetration of the casing.

- (f) Filters
  - (i) All filters shall have permanent frames with replaceable filter media.
  - (ii) The primary filter shall be disposable panel type filter with average arrestance exceeding 90%, dust spot efficiency not less than 40% and initial pressure drop not more than 25 Pa at 0.25 m/s face velocity, filter housed in factory supplied filter frames in accordance with ASHRAE and European Standard EN 1822 Test Standard. Minimum Efficiency Reporting Value (MERV) >6.
  - (iii) The secondary filters shall be disposal panel type mini plead filter with 90 to 95% average dust spot efficiency in accordance to with ASHRAE 52.2 latest edition, and initial pressure drop not more than 170 Pa at 2.5 m/s face velocity, filter housed in factory supplied filter frames in accordance with ASHRAE and European Standard EN 1822 Test Standard. Minimum Efficiency Reporting Value (MERV) 14.
  - (iv) All the filters shall be uniform standard sizes available in the market for future.

#### **G7.1.13 Fan Coil Units**

- (a) General
  - (i) Unless otherwise agreed by the SO, ensure all of the Works comply with the relevant requirements of the Standards and Codes listed below or referenced in the body of the Specification.
  - (ii) BS 4856-4: Methods for testing and rating fan coil units, unit heaters and unit coolers (With and without additional ducting).
  - (iii) Ensure fan coil units are quiet in operation.
- (b) Fabrication
  - (i) Fabricate units of aluminium section framework and thermally/acoustically lined side, bottom and top casing panels constructed to provide a rigid box formation.
  - (ii) Design casings to prevent unauthorised access.
  - (iii) Construct units from 1.6mm minimum thickness galvanized sheet steel rated corrosion- resistant for unit life.
  - (iv) Provide access panels for ease of maintenance of all components.
  - (v) Ensure all components are readily accessible for inspection, maintenance and service without special tools and without disconnection of pipework.
  - (vi) Where required provide a (filtered) outside air spigot connection.
  - (vii) Ensure units are entirely suitable for fixing into the positions required.
  - (viii) Provide double-bottom drip trays under cooling coils, valves and uninsulated internal piping of sufficient size to receive condensation drips from any part of the unit.
  - (ix) Make drain pans at least 25mm deep and fully draining, with a trapped outlet.

- (x) Provide insulated double-bottom drain pans of sufficient size to receive condensation drips from any part of the unit and including all header bends and associated valves/fittings at the inlet/outlet of the pipe connections. Insulate drain pans externally with at least 25mm fibreglass insulation minimum of 48 kg/m<sup>3</sup> density faced with double-sided aluminium foil to prevent condensation.
- (xi) Use board of 32kg/cu.m minimum density.
- (c) Air Filters
  - (i) Provide lay-in replaceable panel air filters in the recirculated air inlet and shall be type at least 25mm thick.
- (d) Air Coils
  - (i) Provide chilled water cooling coils mounted above a drain pan sized to include for collection from the automatic control valve.
  - (ii) Use copper tube, aluminium fin type with 560 fins/metre maximum.
  - (iii) Restrict face velocity at design airflow to 2.5m/sec.
  - (iv) Provide suspension attachments fixed to the unit framing.
  - (v) Reinforce casings local to suspension points. Do not leave casings with exposed edges and gaps at joints or butting sections. Repair cut edges with zinc-rich paint.
  - (vi) Line piping penetrations to casing with rubber sleeves.
  - (vii) Provide finish to casing as specified.
  - (viii) Fabricate drain pans from galvanized mild steel with an epoxy resin paint finish on internal surfaces. Coat underside with 3mm minimum thickness fireproofing material, spray or brush applied.
- (e) Fan sets
  - (i) Provide statically and dynamically balanced forward curved double inlet double width centrifugal type fans with totally enclosed fan ventilated single phase motors with sealed-for-life bearings, requiring no routine maintenance
  - (ii) Provide motors suitable for a 230V 1ph. 50Hz electrical supply.
  - (iii) Restrict motor speed to 24 revs/sec (1450rpm)
  - (iv) Install fans and motors on an independent fabricated common mounting plate.
  - (v) Fix assembly to the unit framework with resilient mountings to eliminate vibration transmission.
- (f) Installation
  - (i) Protect complete fan coil unit during transportation and storage until ready for installation.
  - (ii) Protect fins and casings of the units from damage during installation.
  - (iii) Cover all pipe and duct openings and outlets until final connections are made.
  - (iv) Fit floor units onto steel-spring telescopic anti-vibration mountings with built-in level bolts and neoprene sponge stabilizers.

- (v) Support suspended units on steel-spring hangars with neoprene acoustic isolation washers.
- (vi) Provide drain extension to the nearest floor trap or floor waste.
- (vii) Coordinate with the Plumbing and Sanitary Installation sub-contractor on the location and extent of insulation required for the floor wastes and floor traps; insulation for the affected floor wastes and floor traps shall be provided up to the vertical waste stacks

#### **G7.1.14 Ductwork Thermal Insulation**

Ensure that insulating materials are new, and applied as required within the standards and codes listed and manufacturer's recommended procedures.

Where specified, provide man-made mineral fibrous materials of long fibre classification.

Provide materials to comply with the requirements of BS EN 13166:2012+A2:2016 and BS 3958-5.

Do not use wire netting reinforcement/containment in contact with stainless steel.

Ensure that instruments, gauges and isolating cocks, other fittings and connections are clear of the insulation.

Insulation thickness shall be minimum 50mm or above in non-air conditioned spaces and 25mm or above inside air conditioned spaces.

##### **(a) Excluded Materials**

Do not use the following materials:

- (i) Materials containing mineral fibres not contained or stabilized to prevent fibre
- (ii) migration.
- (iii) Materials using CFCs and HCFCs in their manufacture.
- (iv) Materials containing asbestos.
- (v) Polystyrene.
- (vi) Self-adhesive 'soft' tapes which do not ensure satisfactory long term adhesion.

##### **(b) Adhesives and Mastic Sealants**

- (i) Use adhesives and mastic sealants selected to suit the applications and the exposure
- (ii) conditions that occur.
- (iii) Restrict the use of solvent-based adhesives on site as far as is practicable.
- (iv) Ensure that storage and precautions in use of highly flammable materials complies
- (v) with the recommendations of the manufacturers.

##### **(c) Fibrous Insulation**

- (i) To be to BS 3958-4 with minimum nominal density and maximum A values, at a mean temperature of 50°C, as listed below.

| Description                              | Physical Properties                                                |
|------------------------------------------|--------------------------------------------------------------------|
| Rigid duct insulation                    | 45kg/m <sup>3</sup><br>$\lambda = 0.04\text{W/m K}$                |
| Lamella mat                              | 33kg/m <sup>3</sup> (min.)<br>$\lambda = 0.043 - 0.51\text{W/m K}$ |
| Fibre mattress (medium density)          | 90kg/m <sup>3</sup><br>$\lambda = 0.037\text{W/m K}$               |
| Wire-reinforced mattress (heavy density) | 130kg/m <sup>3</sup><br>$\lambda = 0.037\text{W/m K}$              |
| Duct insulating wrap                     | 45kg/m <sup>3</sup><br>$\lambda = 0.04\text{W/m K}$                |

(d) Phenolic Foam

- (i) Provide phenolic foam preformed insulation to BS 13166:2012+A2, free of water-soluble chlorides, with factory-applied bore coating.
- (ii) Ensure physical properties are as listed below:

|                            |                                                                 |
|----------------------------|-----------------------------------------------------------------|
| Maximum System Temperature | 120°C                                                           |
| $\lambda$ Value            | 0.018W/mK (mean temperature 10°C)<br>0.021W/mK (for ODP = zero) |
| Density                    | 35kg/m <sup>3</sup> minimum                                     |
| Surface spread of flame    | Class O, BS 476-7                                               |
| Ignitability               | Class P, BS 476-12                                              |
| Fire propagation           | I less than 12, i less than 6, BS 476-6                         |
| Smoke obscuration          | Less than 5%, BS EN ISO 5659-2                                  |

(e) Flexible Closed Cell Insulation

- (i) Ensure the insulating material is expanded nitrile rubber elastomeric with Class "O" rating, incorporating a smooth external impermeable barrier surface.
- (ii) Obtain confirmation from the manufacturer for the maximum system temperature.
- (iii) Ensure physical properties are as listed below:

|                                    |                                   |
|------------------------------------|-----------------------------------|
| Maximum System Temperature         | 150°C                             |
| $\lambda$ value (mean temperature) | 0.037W/mK (20°C) 0.039W/mK (40°C) |
| Density                            | 90kg/m <sup>3</sup> minimum       |
| Surface spread of flame            | Class O, BS 476-7                 |



|                                |                                        |
|--------------------------------|----------------------------------------|
| Fire propagation               | I less than 12, i less than 6, BS476-6 |
| Water vapour permeance         | 0.01g/s MN                             |
| Water absorption (by volume)   | 2.5% (28 days)                         |
| Ozone resistance (ASTM D1171)  | 'no cracking'                          |
| Resistance to oils and greases | 'high'                                 |

- (iv) Ensure that calcium silicate complies with the physical requirements of type 1 material to BS 3958-2, and moisture content does not exceed 7.5% when conditioned at medium humidity to BS EN 13468.
- (f) Vapour Barriers
  - (i) Provide vapour barriers of required permeance for the fluid temperature, continuous, and sealed to prevent the passage of water vapour for all surfaces below ambient dew point.
  - (ii) Generally, ensure that the permeance of vapour barriers is to 5.7 of BS 5422.
- (g) Insulation Finishes
  - (i) Shape finishes to suit the installation, with 40mm minimum overlaps at joints arranged to shed water.
- (h) Finish Materials
  - (i) Ensure thicknesses are in accordance with BS 5970, Table 7.

|                               |                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Rigitized Aluminium</b>    | Aluminium sheet embossed or reeded                                                                                        |
| <b>Bright Aluminium Sheet</b> | Aluminium sheets fixed with rivets                                                                                        |
| <b>Stainless Steel Sheet</b>  | Sheet to BS EN 10088-2 or equivalent Standard                                                                             |
| <b>Galvanized Steel Sheet</b> | Galvanized steel sheets with cut edges cleaned and zinc-rich painted; fixed with rivets                                   |
| <b>Aluminised Steel Sheet</b> | Fixed with rivets                                                                                                         |
| <b>Semi Rigid PVC</b>         | Proprietary PVC finishing sheet suitable for tape, solvent weld or plastics rivets fixing, and formed covers for fittings |

- (ii) Apply rigidized, embossed aluminium sheet to all insulated pipework, ductwork and equipment, exposed to possibility of mechanical damage.

#### **G7.1.15 Air-Cooled Split Unit**

Each air-cooled split air conditioning system, shall comprise an outdoor air-cooled condensing unit coupled to single or multiple direct expansion fan coil units as shown in the Drawings. Each unit shall be equipped with suction, discharge and liquid shut off valves with pressure connection for servicing. Factory installed gauge package shall be provided with individual suction, discharge and oil pressure gauges for each compressor supplied with individual shut-off valves. The factory installation shall include mounting and piping to the system.

(a) Anti-Corrosion Treatment

- (i) Anti-corrosion tests shall be conducted by the equipment manufacturer and compliant to accelerated test of ASTM B117 of 3000 hours salt spray test and DIN 50018-2013 of 15 cycles moist Sulphur dioxide test.
- (ii) The outdoor units shall include factory applied heavy anti-corrosion treatment which includes coating to the heat exchanger fins (condenser coil and evaporator coil), bottom frame assembly, mounting plate assembly and partition plate assembly. The treatment shall come with a base coat and top coat of  $\geq 50\mu$ . Coating material for the outdoor unit shall include a primer base coat and top coat of polyurethane or enamel. Units shall come with SUS410 self stainless steel self- screws, nuts and bolts.
- (iii) The Indoor units shall include a primer base coat and top coat of polyurethane or epoxy. The treatment to the heat exchanger fins (condenser coil and evaporator coil) shall come with a base coat and top coat of  $\geq 50\mu$ .

(b) Air-cooled Condensing Units

- (i) Air-cooled condensing units shall be of the packaged type, fully weatherproof, suitable for outdoor installation.
- (ii) All units shall be completely factory assembled, piped, wired, tested and delivered to site complete with all starting gear, controls, instructions and safety devices.
- (iii) Each condensing unit shall comprise compressor, condensing coil, condenser fans, motors, crankcase heater, starter, controls, valves and all necessary auxiliaries. Liquid line solenoid valves shall be provided for condensing units with multiple-step unloading for proper oil return, based on manufacturer's recommendation.

1) Compressor Assembly

- Compressors shall be of the hermetically sealed, welded-shell type and are to be factory dehydrated, sealed and tested. Compressors shall be complete with control and protective devices such as low and high pressure cut-outs, crankcase heater, suction line accumulator, motor winding thermostat and over-current devices.

- The compressors shall be automatically sealed and refrigerant cooled. Each compressor shall have the automatically reversible oil pump. Each compressor shall unload in response to suction pressure down to 50% of full capacity in 2 steps for partial load operation. Multiple compressor units shall have step-start fans and coils.
- 2) Condenser Assembly
- The air-cooled condenser shall be constructed with copper tubes mechanically bonded to aluminium fins to form a cross fin coil. The coils shall be degreased internally and externally before being brazed with Nitrogen passing through the tubes.
  - The condenser coil shall be coated with Base Coat (Primer) + Top Coat (Polyurethane) heavy anti-corrosion treatment of  $\geq 50\mu$ .
  - The coils shall be tested in factory for leaks at 2240 kPa before assembly. They shall again be pressure tested and leak tested after assembly at site.
  - Each fan coil unit shall consist of direct-expansion evaporator coil, blower fan and motor enclosed in a water-proof casing.
- 3) Condenser Fan and Motor
- The condenser fans shall be of the centrifugal direct drive type, with side discharge for units located in M & E plant rooms; propeller direct drive type with vertical discharge for units located in roofs or outdoor spaces.
  - The fans shall be protected with heavy gauge galvanised wire guard, and painted with oven baked polyester paint. The fans shall be statically and dynamically balanced.
  - Fan motors shall be permanent-split-capacitor type, complete with thermal protection and permanently lubricated ball bearings.
- 4) Protective Devices
- The protective devices shall include, and not limited to the following:
- High Pressure Cut-Out;
  - Low Pressure Cut-Out;
  - Fusible Pressure Relief Plug;
  - Compressor Motor Solid State Protection;
  - Five Minute Timed Off, Anti-recycle Timer;
  - Separate System On-Off Switches on each Compressor;
  - Manual Reset Oil Pressure Protection;
  - Internal Compressor Relief Valve;
  - Internally Protected Condenser Fan Motors;
  - Suction Line Accumulator; and

- Separately Fused Condenser Fan Motor and Control Circuit.
- (c) Evaporator Unit (Fan Coil Unit)
- (i) Evaporator
    - 1) The evaporator coil shall be constructed from strong clean copper tubes bonded to aluminium fins suitably spaced to ensure maximum heat transfer.
    - 2) The evaporator coil shall be coated with Base Coat (Primer) + Top Coat (Polyurethane) heavy anti-corrosion treatment of  $\geq 50\mu$ .
    - 3) The inlet of the coil shall be factory brazed to an automatic thermostatic expansion valve. The face velocity of the coil shall be low to ensure quiet operation.
  - (ii) Evaporator Fan
    - 1) The fan shall be a centrifugal blower driven by a permanently lubricated motor. The fan is required to be silent and free of vibration during operation. The fan speed shall be variable by means of a Hi-Med-Lo switch. In general, the sound pressure level at 1 m away from the fan coil shall not exceed 54 dB(A) at 1m.
  - (iii) Filter
    - 1) Panel filters of the washable type with an average arrestance of 85% when tested in accordance with ASHRAE 52.2 shall be provided.
  - (iv) Drain Line
    - 1) There shall be a provision for draining the base pan through a PVC drainpipe to the nearest floor trap, whether indicated in the Drawings or otherwise. Panels shall be easily removable for servicing and maintenance. The drain line shall be insulated with minimum 25mm thick closed cell insulation. All piping shall be supported to prevent buckling at appropriate interval not exceeding 2m.
  - (v) Control System
    - 1) Each fan coil shall be equipped with controls and thermostat for regulating the room temperature. The thermostat of each fan coil shall control its respective solenoid valve on the refrigerant cycle. Only system designed and manufactured in the factory are acceptable.
    - 2) The main features of the control panel are:
      - Cooling and fan operation;
      - On/off timer function;
      - Timer remaining time display;
      - Test run function; and
      - Serving louvre selector.

(vi) Thermostat

- 1) A thermostat of an approved type shall be provided. In the case where there are two compressors in the condensing unit, the thermostat shall be of the two stages type. The thermostat shall be of the on/off type with an operating range of 15degC – 30degC with a differential of 1degC suitably mounted in the room to sense the return air temperature, to cycle the compressor for temperature control. A built-in thermostat shall be incorporated. This thermostat shall be designed to operate on 24 VAC electrical supply. The wall mounted and ceiling suspended fan-coils shall be provided with remote control switch-cum-thermostat with 3m of control wiring.

(vii) Noise levels in the rooms

- 1) It shall be the Contractor's responsibility to ensure that the equipment chosen shall not induce a noise level greater than specified in the rooms. If necessary, acoustic treatment shall be provided.

(viii) Refrigerant Piping/Control Wiring

- 1) All refrigerant piping shall be of hard tempered, seamless copper tubing to BS EN 12735-1.
- 2) Pipe sizes shall be adequate to pass the refrigerant quantities required at normal velocities with the pressure drop commensurate with the available head. Refrigerant pipe sizes shall give pressure drops of not more than 20.7 kPa (3 psi) between compressor and condenser and 10.34 kPa (1-1/2 psi) between evaporator and compressor. Velocities shall be adjusted to ensure safe carriage of oil under operating conditions.
- 3) The whole of the refrigeration pipe lines including gas suction line, all fittings, valve and strainer bodies, and flanges on the refrigeration equipment shall be insulated with minimum 25mm thick closed cell insulation. The insulation shall be wrapped with a double-sided vapour barrier sheet and encased in 24 gauge aluminium sheathing held in position by means of 25mm wide x 24 gauge aluminium iron strips at 900mm centres.
- 4) The refrigerant piping for each refrigerant circuit shall include a filter drier, a moisture indicating sight glass and services access valves.
- 5) All control wiring for the installation shall be run in conduits.
- 6) All refrigerant piping and electrical wiring in conduits shall be run in PVC trunking. The trunking shall be properly secured to the wall above the false ceiling by means of suitable hot dipped galvanised brackets. All trunking exposed to room space shall be embedded into the wall. All trunking subjected to ingress of water shall be sealed tight with approved sealing compound.
- 7) In the outdoor area, the insulated refrigerant pipes and control wiring shall be further enclosed with in PVC trunking of UV-resistant type or SS 306 material.

#### **G7.1.16 Variable Refrigerant Volume (VRV) systems**

Each VRV system shall comprise an outdoor air-cooled condensing unit coupled to multiple fan coil units as shown in the Schematic Drawings. All units shall be completely factory assembled, piped, wired, tested and delivered to site complete with all starting gear, controls, instructions and safety devices.

(a) Anti-Corrosion Treatment

- (i) Anti-corrosion tests shall be conducted by the equipment manufacturer and compliant to accelerated test of ASTM B117 of 3000 hours salt spray test and DIN 50018-2013 of 15 cycles moist Sulphur dioxide test.
- (ii) The outdoor units shall include factory applied heavy anti-corrosion treatment which includes coating to the heat exchanger fins (condenser coil and evaporator coil), bottom frame assembly, mounting plate assembly and partition plate assembly. The treatment shall come with a base coat and top coat of  $\geq 50\mu$ . Coating material for the outdoor unit shall include a primer base coat and top coat of polyurethane or enamel. Units shall come with SUS410 self stainless steel self-screws, nuts and bolts.
- (iii) The Indoor units shall include a primer base coat and top coat of polyurethane or epoxy. The treatment to the heat exchanger fins (condenser coil and evaporator coil) shall come with a base coat and top coat of  $\geq 50\mu$ .

(b) Condensing Unit

- (i) All condensing unit shall be package type, fully weatherproof, suitable for outdoor application. The package unit shall be located and arranged such that discharge of hot air does not cause a nuisance and recirculation to the fresh air intake is avoided.
- (ii) Condensing unit shall be designed such that the noise level will not exceed 57 dB(A) when measured 1m away horizontally and 1.5m above ground.
- (iii) Condensing unit shall be provided with inverter type compressor of the high efficiency and good energy consumption type complete with IC control system to ensure reliability. Each condensing unit shall be of the hermetic reciprocating compressors, each suitable for satisfactory operation down to 50% of its rated capacity. The refrigerant circuit shall be complete with an accumulator, oil recovery system, liquid and gas shut off valves and a solenoid valve, to ensure safety operation of the system. Use of oil trap shall not be allowed.
- (iv) Each condenser unit shall be of rust resistant outside galvanised steel plate and double-baked acrylic resin finish. The casing enclosure partition plate, mounting plate and bottom frame shall be treated with heavy anti-corrosion treatment with Base Coat (Primer) + Top Coat (Polyurethane)  $\geq 50\mu$ . The casing shall be suitably coated with a bake enamel finished. All self-tapping screws used shall be of the SUS410 stainless steel screws.
- (v) All condensing unit shall be provided with concrete plinth using 8mm diameter steel bar. Concrete used shall have 1:3:6 mix. There shall be a minimum of 4% reinforcing steel by weight. The foot of the equipment shall have a house pad of concrete construction such that the house pad is at least 50mm higher than the general concrete plinth. The detail drawing of the plinth and house pad shall be submitted to the S.O. for approval before commencement of work at site.

- (c) Evaporator Unit (Fan Coil Unit)
  - (i) Fan coil unit shall be of sufficient rigidity. Any panels of the fan coil unit found to be observed to vibrate under operating condition shall be rejected and the Contractor shall be responsible for replacing the unit without inconvenience to the Board. Each fan coil unit shall be equipped with remote IC thermostat sensor, which control room temperature.
  - (ii) Fan coil unit shall consist of electronic control valves, evaporator fan, safety start circuit, freezing up protection thermostat, etc. Evaporator coil shall be copper tubes expanded into aluminium fins. The coils shall be degreased internally and externally before being brazed with Nitrogen passing through the tubes.
  - (iii) The coils shall be tested in factory for leaks at 2240 kPa before assembly. They shall again be pressure tested and leak tested after assembly.
- (d) Thermostat
  - (i) The thermostat sensor shall be located within the fan coil unit without being exposed to view and shall be effective in sensing an accurate return air temperature. The manufacturer shall be required to guarantee the performance of the fan coil unit under this condition.
  - (ii) Control diagram for field wiring to the remote controller shall be provided by the Contractor for the S.O.'s approval before commencement of works. The fan coil unit shall be provided with remote control switch-cum-thermostat with 3 m control wiring.
- (e) Drain Line
  - (i) There shall be a provision for draining the base pan through a PVC drain pipe to the nearest floor trap or vertical rain drain pipe, whether indicated in the Drawings or otherwise. Panels shall be easily removable for servicing and maintenance. The drain line shall be insulated with minimum 25mm thick closed cell insulation. All piping shall be supported to prevent buckling at interval not exceeding 2m.
- (f) Filter
  - (i) Each fan coil unit shall be provided with a washable air filter, of minimum 70% arrestance tested in accordance with ASHREA 52-76.
- (g) NC levels in the Rooms
  - (i) It shall be the Contractor's responsibility to ensure that the equipment chosen shall not induce a noise level greater than specified in the rooms. If necessary, acoustic treatment shall be provided under the Contractor.
- (h) Refrigerant Piping/Control Wiring
  - (i) All refrigerant piping shall be of hard tempered, seamless copper tubing to BS EN 12735-1.

- (ii) The whole of the refrigeration pipe lines including gas suction line, all fittings, valve and strainer bodies, flanges on the refrigeration equipment shall be insulated with minimum 25mm thick closed cell insulation. The insulation shall be wrapped with a double-sided vapour barrier sheet and encased in 24 gauge aluminium iron sheathing held in position by means of 25mm wide x 24 gauge aluminium iron strips at 900mm centres.
- (iii) The refrigerant piping for each refrigerant circuit shall include a filter drier, a moisture indicating sight glass and services access valves.
- (iv) All control wiring for the installation shall be run in conduits.
- (v) All refrigerant piping and electrical wirings in conduits shall be run in PVC trunking. The trunking shall be properly secured to the wall of the false ceiling by means of suitable hot dipped galvanised brackets. All trunking exposed to room space shall be embedded into the wall. All trunking subjected to ingress of water shall be sealed tight with approved sealing compound.
- (vi) In the outdoor area, the insulated refrigerant pipes and control wiring shall be further enclosed with PVC trunking.

**END OF SECTION**